

The Fernstudent.

A User-Centered Design Case Study on Improvements of Hybrid Teaching Through A Physical Avatar in the Classroom.

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ABSTRACT

Hybrid formats have become a common approach in university teaching. Typically, there is a synchronous live session which students can join either in the physical classroom or digitally/remotely with interaction through textual chat. While this has many advantages in terms of inclusivity and sustainability, it also poses problems such as increased attention requirements for teachers and a lacking social integration of remote and on-site students. Based on a user-centered design approach, we developed the concept of the "Fernstudent". A physical avatar represents remote students collectively in the classroom, creating a communication channel based on the same modalities as for on-site communication. The present case study describes the iterative design process, from the first idea via explorative interviews and state of the art analysis, concept development, potential analysis, exploration of different design variants, a first functional prototype, a pilot field test, and finally lessons learnt and planned future steps of development.

CCS CONCEPTS

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing devices.

KEYWORDS

Fernstudent, hybrid teaching, user-centered design, concept development, prototype, pilot field test

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1 INTRODUCTION

After the (enforced) intense experiences with digital and hybrid learning in the course of the COVID-19 pandemic, now, coming back to the "regular" mode of university life, the question for us as researchers and teachers at a university is what we might want to

carry over. For example, do we still want to offer a hybrid option for lectures?

While the term hybrid teaching is not used consistently, and may refer to different ways of integrating technology in education (e.g., [8, 11, 17]), it usually refers to a combination of on-site and digital teaching in a synchronous live-session, where remote students follow the lecture via video and audio channels using an online platform (e.g., Zoom, Jitsi Meet), with the opportunity to pose questions or comments via textual or voice chat (e.g., [12]). Informal discussions with colleagues showed a split opinion on hybrid teaching: while some saw this as a worthwhile option and positive signal of progress and inclusion which would also be expected from a modern university, others clearly declared they would not offer a hybrid option, because of the many disadvantages. In particular, these others complained about the additional demands to handle the lecture's slides and the online platform to integrate remote students on one screen at the same time, which would definitely impair their attention and teaching quality. Also, the splitting of the students' group in on-site and remote would impair the quality of interaction. In the end, one teacher declared that they can only focus on one place and one group, so that the other group is left out. As HCI researchers, this made us think: Does it really have to be like this? Or could there be a technical solution, that would allow teachers to focus on one place, but still include remote students? This was the birth of the idea of the "Fernstudent" (German compound word, originally referring to students at a remote university). Driven by the aim to improve the quality of hybrid teaching and make it a worthwhile option for both, teachers and students, we received funding from our university's Quality Teaching Fund to develop corresponding concepts in a user-centered design approach. While we started our exploration at our own university and, thus, focused on the needs and experiences of teachers and students at LMU as informants, we also aim to disseminate the concept and possibly adapt it to other universities and meeting contexts in future work.

The present case study describes the user-centered iterative design process, from the first idea via explorative interviews and an analysis of the state of the art, concept development, potential analysis, exploration of different design variants, a first functional prototype, a pilot field test, and finally lessons learnt and future steps of development. While insights from the user studies and relevant backgrounds from educational psychology are reported in more detail elsewhere [13, 14], the present case study focuses on the HCI practice and translation of user needs into concrete technical design decisions, the challenges and solutions, impacts on real-world teaching, and implications for HCI research and design, also beyond the context of teaching and the university.

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2 EXPLORATIVE INTERVIEWS AND STATE OF THE ART ANALYSIS

Explorative interviews with two university teachers and 57 students provided us deeper insights into the particular needs and experiences of these two relevant groups in the context of hybrid teaching (also see [13]). The interviews were structured along three leading questions, i.e., (1) What are your (positive and negative) experiences with hybrid teaching? (2) What are the advantages and specific qualities of face-to-face compared to hybrid teaching? (3) What should be changed to make hybrid teaching an (even) better option? Do you have any ideas on how technology could support this? From the teacher's perspective, hybrid teaching was mostly experienced as one-directional communication, resulting in a bad atmosphere and lacking a feeling of connectedness. Also, hybrid teaching was described as "a bad compromise", where the fact that it is live would create more stress, but the advantage of being live was hardly taken advantage of. Both, teachers and students also emphasized the lacking connection between on-site and remote students. Furthermore, the students complained that the teachers felt overburdened and some teachers would even explicitly state that questions from remote students were not allowed. On the other side, the teachers complained that they didn't get anything back from the remote students and would feel uncomfortable, especially if remote students turn off their camera ("... the ones who are remote might as well be gone [...] you are totally exposed but speaking to people you can't see").

In the end, both, teachers and students, emphasized the necessity to bring both student groups (on-site and remote) more on one level (in visibility, modality, social interaction, appearance in the room) and provided first design ideas, such as a loudspeaker for questions from remote students or a microphone that records not only the teacher but also the audience, so that remote students could better follow discussions in the physical classroom.

A state-of-the-art analysis of literature on hybrid teaching revealed a similar picture. The main advantage is its inclusive character, i.e., even those who are unable to be present in the physical classroom, for example for reasons of childcare, illness or infection risks, or traveling (e.g., [5, 6]), still can attend a lecture. On the other hand, the format comes with numerous difficulties, connected to impairments of wellbeing and educational quality (e.g., [3, 4, 9]). As studies showed, both parties (remote and on-site) show behavior that contributes to growing psychological distance and reduced interaction between the two settings. Remote students often engage in side-activities while attending their lecture and participate less actively than on-site students [9]. The teachers, in turn, tend to focus their attention on the on-site students, and do this even more if questions or comments from remote students are rare [4]. Overall, experiences reveal two primary objectives for the improvement of hybrid teaching: (1) to reduce the cognitive load on the part of the instructor and (2) to reduce the differences in social presence and interaction opportunities between remote and physically present students.

A straightforward approach to enhance the social presence of remote students in the classroom could be a physical avatar, which serves as a prominent "reminder" of the remote students joining the lecture while naturally supporting a style of interaction with a

social entity (see also [1, 2]). This approach also has some similarities with the idea of telepresence robots (TPRs), as already applied in diverse educational settings (e.g., [7, 15, 16]). A telepresence robot is typically a 1:1 representation of a remote person, usually displayed on a screen, through which the remote person can operate the camera angle to control their view at the on-site setting (for examples see <https://telepresencerobots.com>). Compared to this, the present approach is an adoption of the TPR tailored to the university context. Instead of a 1:1-representation, all remote students are represented by a single physical entity placed at a fixed spot among the on-site students (which seems more practical and affordable than one TPR for each remote student).

3 CONCEPT DEVELOPMENT

Based on these insights about the key requirements, in an internal workshop, we sketched a preliminary first concept (called the "Fernstudent") with the following basic features: A **physical avatar** with a physical presence, "sitting" in the lecture hall, side by side with the on-site students, representing the remote students, whereby **image and sound from the lecture hall are transmitted to the remote students**, and **images and sound from remote students are transmitted into the lecture hall**. **Remote students can indicate that they want to speak** (e.g., by motor elements, "hand signals", light signals) and the **number of remote students is somehow displayed in real time**. With these features, the concept of the Fernstudent addresses the main identified challenges of hybrid teaching: (1) It reduces the cognitive demands for the teaching person by limiting the focus of attention to one spot and modality. Signals from the remote students become visible in the same space as those from the on-site students, which frees up the instructor's attention capacities for lecture content and interaction quality. (2) It enhances the presence of remote students by a physical, anthropomorphic representation and places both student groups on a more equal level (in visibility, modality, and opportunities for social interaction).

4 POTENTIAL ANALYSIS: CONCEPT EVALUATION WITH TEACHERS

Before we went on with a further realization of the concept as a functional prototype, we presented the concept (by means of a short description and scribbles) to a convenience sample of 21 teachers at our university from diverse departments (e.g., politics, literature, law, computer science, economics) and collected their spontaneous opinion about the potential of the concept. Before presenting the concept of the Fernstudent, the teachers reported about their experiences with hybrid teaching in general and their personally preferred teaching mode (e.g., on-site, online, hybrid). Each interview lasted about 30 minutes. The interviews were transcribed verbatim and statements were broadly categorized. The following sections report the central results. Regarding the question of preferred teaching format, the largest group had a clear preference for the classical on-site teaching (9 of 21, 43%). Five teachers (24%) declared that depending on the type of course (e.g., lecture, seminar, group work) both, *on-site* and *online* could be preferable, but not a *hybrid* solution. Four teachers (19%) opted for hybrid teaching, but stated that remote participation should only be a back-up solution and

on-site participation should remain the standard. Three teachers (14%) were unsure. After a short introduction to the concept of the Fernstudent, teachers reflected on the imagined mental load with either a hybrid lecture with the Fernstudent in the classroom or a classical hybrid format, using a software such as Zoom or Jitsi Meet. The majority (17 of 21, 81%) rated the mental load higher for the classical hybrid format. One (5%) rated it higher for the Fernstudent, "because it is something that I am not used to, simply because it is new" [P6]. Three teachers (14%) thought that this was hard to tell or probably similar for both.

Regarding the functionality and modalities of interaction, the vast majority (18 of 21, 86%) stated that remote students should be able to pose questions or comments via speech, using their natural voice (just like the on-site students). Two teachers (10%) suggested that a robot voice should read out what the remote students typed in via chat. One teacher (5%) found that it doesn't matter whose voice it is, as long as it is understandable. With regards to a video option (i.e., the teacher can not only hear but also see the remote students), 14 teachers (67%) suggested that all participating remote students should be visible with their face, because it is a "better feeling if you see who you are talking to" [P10], and also explaining that the mimic expressions are an important (nonverbal) feedback for the teacher, where they see that something might have to be explained in more detail [P10, P17, P23]. Four teachers (19%) declared that it was not necessary for them to see the remote students, and 2 teachers (10%) said that at least the speaking student should be visible (1 person was not sure).

Sixteen teachers (76%) clearly wished for a feature that displays the number of remote participants. Two teachers (10%) stated that this would be a nice feature but only if it is easily recognizable, and that the exact number of participating students was not important. Three teachers (14%) were not sure. Finally, regarding the design of the avatar, there was no clear opinion among the interviewed teachers. One third (7, 33%) thought that it should have a human-like design, whereas another third (7, 33%) opted for a minimalistic, non-human like design. Another third (7, 33%) declared no preference and emphasized that the focus was on the functions and not the design.

In sum, the potential analysis thus confirmed the Fernstudent as a promising concept, worth to be further developed and tested. The majority of teachers assumed that this could reduce their mental load. Also, the teachers' opinion confirmed the planned features such as a display of the number of remote participants and video and audio transfer between classroom and remote students. Regarding the latter, we also discussed that the transfer of the video of the remote students could be an optional feature and it should be decided jointly in a course how to deal with this (e.g., an agreement that all switch the camera on or all off). Regarding the design of the avatar, we came to the conclusion that it could have human-like elements but altogether should have a neutral, minimalistic impression that keeps the focus on the remote students which it represents instead of becoming a focus of attention itself.

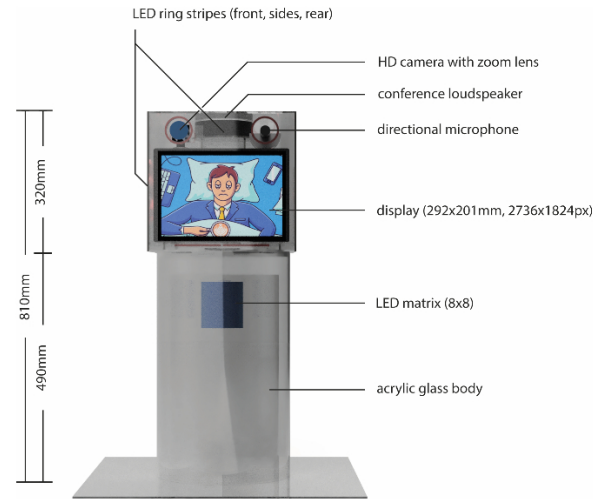


Figure 1: Set-up of hardware components.

5 TECHNICAL REALIZATION AND PROTOTYPE

In order to implement the Fernstudent as a functional prototype, we had to address two basic questions: the technical realization, including hardware and software, and its physical shape and aesthetic appearance. In the following sections we describe how we came to the current solutions for each of these issues.

5.1 Technical realization

5.1.1 Hardware. Figure 1 shows the set-up of hardware components, which are described in more detail below. In order to reduce complexity and keep space requirements to a minimum, we wanted to include as few devices as possible. Thus, we started from a tablet computer with an integrated touch interface, so that there no additional keyboard is needed and all commands can be realized via the screen. We used a **Microsoft Surface 7+ notebook** which also features a front camera and microphone.

Video transfer. Our first idea was to use the integrated notebook camera to record the teacher. However, first tests showed that the camera angle was too wide which meant that the notebook had to be positioned fairly close to the teacher and projector image. As we did not want the position of the Fernstudent in the classroom to be determined by the camera angle, we decided for an **external camera** with a zoom lens (Raspberry Pi Zero with HQ Camera Module, 8-40mm Zoom). This worked well to record the teacher, however, it turned out that the light sensitivity was not always sufficient to capture the sometimes faint image of the lecture slides. Hence, in order to transfer the slide image safely to the remote students, we used a **video capture card** (Elgato HD60 X) that loops the signal through to the projector. In the end, the remote students see a combination of the captured image and the filmed teaching person, which resolves all lighting issues.

Audio transfer. Our requirement was to transfer audio signals from the classroom to the remote students, so that they can follow the lecture and discussions on-site, and to transfer audio signals

from the remote students to the classroom, if one of the remote students wants to pose a question or join the discussion. Regarding the transfer of audio signals from the classroom to the remote students, we needed a microphone that can transfer near and far sounds. The notebook's built-in microphone was too insensitive and a dedicated microphone worn by the teacher was not an option because we wanted all technology to be integrated in the Fernstudent. We therefore used a **directional microphone** (Rode VideoMic GO II), that primarily records the teacher. Nevertheless, if the teacher does not say anything, the microphone becomes more sensitive to other sounds which are then also transmitted. Hence, also the on-site students' questions and comments are transmitted to the remote students. This has the advantage that the teacher does not have to repeat questions from the audience, and it promotes a natural flow of discussion. However, a disadvantage is that if for some reason the teacher does not say anything (e.g., during a silent exercise), every noise from the classroom is transmitted, even paper rustling. Regarding the transmission of audio signals from the remote students to the classroom, the Fernstudent prototype was equipped with an **external conference loudspeaker** (Jabra Speak2 75).

Signaling function. As a signal that one of the remote students wants to say something, we decided for a light signal (prior ideas also referred to the Fernstudent lifting its "arm", however, this would have required a more human-like shape of the prototype and more complex requirements regarding motor elements, which we decided against). The light signal was realized via a set of LED ring stripes controlled by a microcontroller (Arduino Uno). When one of the remote students clicks the "raise hand" icon, a **LED light circle, signaling that one of the remote students wants to say something** illuminates. The LEDs were chosen so that brightness and color could be adjusted to reach an optimal level of salience. In parallel to one of the on-site students raising one's hand, this is a decent signal which is clearly visible for the teacher but also not too distracting or annoying (like, for example, a flashing light) so that it would disturb the ongoing lecture. After the remote student has spoken, they press the "lower hand" icon and the light circle vanishes. In addition, another **LED matrix shows the number of dialed-in remote students**.

5.1.2 Software. Figure 2 shows the communication diagram of the software components as described in more detail in the following sections. We searched for an existing software such as a videoconferencing tool with an integrated raise hand function that can be intercepted in order to be able to translate it into the light signal. This was not possible for many existing videoconferencing tools (e.g., Zoom, that removed the feature in its API). A possible option was Jitsi Meet, which is an open software in which this could be realized. Hence, we used **Jitsi Meet as a software** and set up a server at our university that runs the **Jitsi server**. For the remote students the process of joining a lecture is similar as with other meeting tools. They simply follow a link in their preferred browser, choose a name and are ready to join without the need for additional programs or apps. In addition, the Fernstudent needs to join the meeting as well (which the teacher or some other on-site person has to set-up before the lecture starts). The remote students then see a merged image combining the teaching person and the lecture

slides, which is realized via **Open Broadcaster Software** (OBS, <https://obsproject.com/de>).

The required interactivity was realized via a **React App** with **Node.js** (<https://react.dev>; <https://nodejs.org>), which runs on the Fernstudent computer and inserts the Fernstudent as a participant in the Jitsi meeting. Furthermore, the app monitors Jitsi events, such as raised hands or the joining or leaving of students, which results in a change of the total number of remote students. Those events are then transmitted to the Arduino, which, in turn, initiates actions such as turning on the LED ring stripes or update the number in the LED matrix. Similarly, the Arduino captures button touches on the remote control (such as *reset raised hands*) and sends them to the React App, which, in turn, sends commands to the Jitsi Server. Thus, we have the Jitsi-Server on the one side, Arduino on the other side, and the app mediates between these two.

5.2 Physical shell and aesthetic appearance

Initially, we experimented with different design variants in a practical course on experience design. Students' ideas ranged from relatively abstract shapes to animal-like shapes or toy figures (see Figure 3). However, informal pre-tests showed that playful designs were not ideal for the "serious" university context. Characteristic variants (e.g., Lil'E) or variants based on existing movie characters (e.g., Dr. Minion) generated more polarized judgments, which we considered inappropriate for an entity that should represent many students at the same time. Thus, we decided for a relatively simple geometric form, i.e., a **cuboid** and a **cylinder** built from **acrylic glass**, which contained the required technology. The cuboid actually is double-walled and contains a second, smaller cuboid for the mounting of the equipment which also hides the technology from outside. We also added cutouts for the camera and directional microphone. The power supply units are collected in the bottom of the cylinder. Regarding size, the basic idea was that the Fernstudent "sitting" in the lecture hall should be on face level with the on-site students. Moreover, the exact size was also determined by the size of the technology inside (e.g., the size of the notebook determined the edge length of the cuboid). Figure 4 shows a smaller prototype from experimenting with different materials and ratios of forms and the final prototype which was then used in a field test.

6 PILOT FIELD TEST

The Fernstudent was applied in multiple sessions of different lectures at our university during the summer semester 2023. The lectures were given by the same teachers who also served as informants about their experiences with remote teaching in the explorative interviews. Students were free to decide whether to join the lecture in presence or remotely. During the test phase, students and teachers filled in a survey after each lecture and rated their experience of the lecture by a combination of quantitative and qualitative measures. In addition, retrospective in-depth interviews were conducted with the two teachers after the end of the semester. Altogether, the test phase lasted nine weeks. In the first weeks, the hybrid option was realized via the regular Jitsi Meet software with no physical avatar, and in the second part of the test phase, the Fernstudent was used for the hybrid option. The Fernstudent and its basic features were introduced in the first test session. The

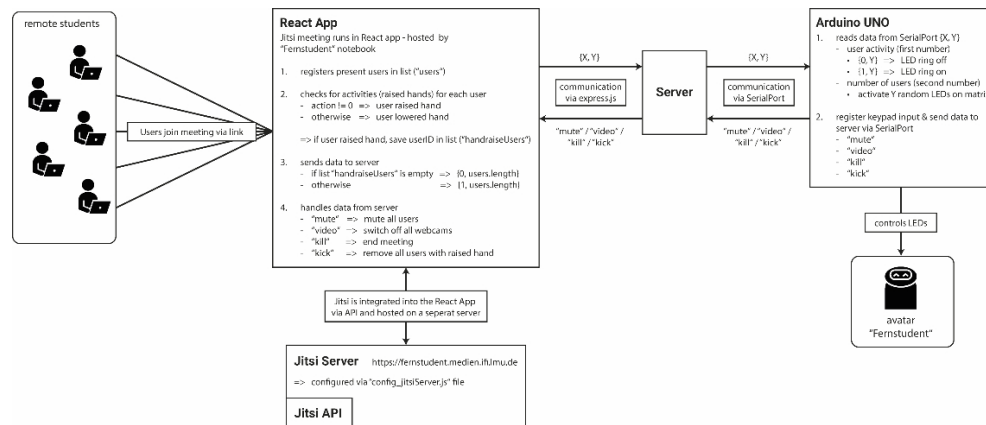


Figure 2: Communication diagram of software components (Figure adapted from [10]).

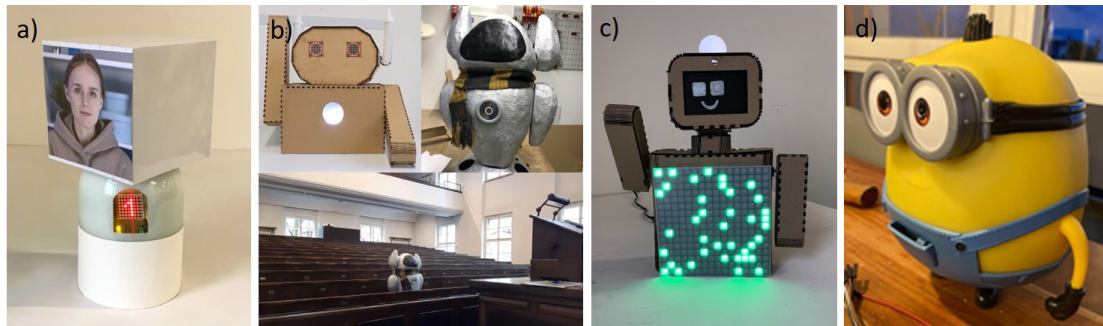


Figure 3: Different design variants created by students of the experience design course (<https://blockpraktikumexperiencedesign.wordpress.com>: a) Student cube, b) Lil'E, c) Remotey, d) Dr. Minion



Figure 4: Fernstudent prototypes: a) Smaller prototype from experimenting with different materials and ratios of forms and b) + c) the final prototype used in a field test.

study was approved by the institutional review board and informed consent was collected from all participants.

In summary, the field test showed promising results both from a teachers' and a students' perspective (for details see [13, 14]), however, different aspects of the concept were of in the focus of attention. From the teachers' perspective, the main benefit of the

Fernstudent was the reduction of stress and cognitive demands, which allowed them to focus on the lecture content and discussion with the students, instead of the handling of technology. Also, they experienced the interaction with the remote students as more satisfying and more natural, and they experienced the two groups of students more like one category [13].

Regarding the students' perspective, the main changes initiated through the Fernstudent refer to the experienced social presence and the affordances for participation for the remote students. More specifically, the survey data showed that compared to the regular hybrid sessions, the perceived social presence of the remote students became higher with the Fernstudent, and the interaction between teacher and remote students was perceived as being more similar to that between teacher and on-site students. Furthermore, remote students reported a higher intention for active participation (raising their hand) in Fernstudent sessions compared to Jitsi Meet, up to a similar level as the on-site students. Regarding the perceived character and impression of the Fernstudent in the classroom, the students described it as mainly neutral and mainly pragmatic. This generally corresponds to our intention of giving the remote students a presence in the classroom but not introducing a technology that becomes a character and source of attraction itself [14].

In sum, the pilot field test of the Fernstudent showed benefits for all groups involved, however, on a different level and regarding different qualities. Among the three relevant user groups (teachers, on-site and remote students), the least changes for the on-site students. For them, the Fernstudent appears as a new element in the classroom and their remote classmates become more present, but it doesn't change their own behavior. They still listen to the lecture and raise their hand as usual. In contrast, for the teachers, the Fernstudent changes a lot, also regarding their behavior and tasks as a teaching person. Compared to the regular hybrid format, they get rid of several tasks which formerly created stress and captured attention. Instead of watching the chat and online-tool in parallel to what is happening in the classroom, with the Fernstudent, signals from the remote students become visible in a more natural way. The focus of attention can remain in the classroom, where signals from on-site and remote students are visible on one level and the interaction takes place in the same modality. In the teacher interviews, this was reflected in reports about stress reduction, a more relaxed teaching session and feeling more competent as a teaching person. For the remote students, in the present form of the Fernstudent, the biggest change refers to the enhanced modalities of interaction (participating in the discussion via speech, better hearing what the on-site students are saying), which probably made them feel more involved in the happening and in turn encouraged a more active participation in the lecture. Also, they are aware of being "more visible" in the classroom, however, their visibility in the classroom is still not much represented in what the remote students see on their screen at home. Amongst other, this issue was picked up for further developments of the concept, as described in the next section.

7 LESSONS LEARNT AND FUTURE STEPS

The encouraging results and feedback from students and teachers in the potential analysis and pilot field test confirmed the general potential of the concept. At the same time, the experiences so far also provide an opportunity for further developments and improvements of the concept. We thus conducted a workshop with four teachers and four students involved in the project. The aim of the

workshop was to reflect on lessons learnt and to discuss reasonable further development of the concept, regarding technical and conceptual issues.

7.1 Technical issues

A recurring issue was the **audio quality**, which was mentioned by teachers and students alike. While the overall audio quality was quite satisfying, there were several instances where communication was hindered. From the remote students' perspective, the microphone's sensitivity adjustments were an advantage in some situations but a disadvantage in others. While it ensured that every student in the classroom could be heard clearly by the remote students, in silent conditions background noise was amplified and annoyed remote students. In future iterations, more fine tuning of the microphone settings or additional features such as a squelch function (to address background noise) could be a starting point for improvement. Another issue was the **practical handling and mobility of the prototype**. With the current prototype, two students were designated to handle the set-up for the field test, i.e., putting together the acrylic frame and the hardware elements, wiring everything together, and setting up the software. Given that the concept aims to provide a seamless solution with low barriers for entry, any complicated set-up process would be contradicting. Therefore, an improvement in practical handling and mobility of the prototype for easier removal between lecture rooms or even universities and other institutions must be addressed in future iterations. A slimmer, potentially foldable design, could be a starting point combined with default settings for lectures, so that teachers just have to open and start the system to be ready to go.

7.2 Further developments of the concept

From the teachers' point of view, there were some starting points for **improving the interaction with remote students** and making the opportunities even more similar to those regarding on-site students. For example, **"hand raises" by remote students could be less anonymous**. In the current iteration, the Fernstudent shows the last remote student which actively used voice chat. While that was sufficient for many situations, in the event of a new hand raise, teachers might like to know who the student is who wants to talk. Another improvement could be an **indication of how many questions are in the queue**. While the present concept works well in smaller groups it could quickly go out of control if several hundred remote students are present. Lastly, an **alternation of the shown remote student** by camera picture would underline that the Fernstudent is no classical telepresence robot and represents not one but many. This would ensure that teachers perceive the diversity of joined remote students including the quiet ones who otherwise remain invisible.

Another discussion revolved around the question of **camera usage as default** or even mandatory. One perspective is the privacy of remote students, who might not want to turn on their camera for a variety of reasons. However, there are some implications to consider, contradicting the core intentions of the concept. First, the core concept of the Fernstudent is the idea of bringing all students, remote and on-site, to the same level, including presence, interaction capabilities, and also visibility. Making the camera optional

would undermine this core idea resulting again in differences between the student groups. Second, turning the camera off would make individual remote students invisible, promoting secondary activities, which again decreases interaction probability. Third, while we are only beginning to understand such hybrid teaching scenarios, depending on the concept features, we can anticipate an unfolding culture of remote students. For example, if many students use the text chat, this could motivate others to do the same. Similarly, if most students have their camera turned off, others will likely do the same, simply to adapt to the existing culture. The result could be that the default is turned-off cameras, a situation witnessed by many teachers when they talk to a “wall of black screens”. How such a culture would evolve is hard to predict but we can use design to make an inclusive culture characterized by increased interaction more likely.

Another issue was the **presence of remote students in the classroom from their own perspective**. While they were physically represented by the avatar among on-site students, and therefore were present from the perspective of teachers and on-site students alike, from their own perspective they only saw the composition image of lecture slides and a smaller video feed of the teacher. Hence, it was hard for them to get really immersed and feel present. We discussed design solutions such as a 360-degrees field of view or adjustable camera angles but dismissed those ideas for privacy reasons and uncertainties of who could be in charge of the camera angle (there are many remote students but only one adjustable angle). We came up with alternative, more promising solutions, which aimed not so much at the remote students’ current perceptions but rather at the conceptual understanding (i.e., mental model). More precisely, we envisioned two different approaches. The first is derived from computer games in which you control a character from their point of view and which sometimes begin with a perspective flight where you see a character from above and then you fly into their body which grants you control over them. Such a **fly-in sequence** could be shown right after logging in where joining remote students see the lecture hall from above, flying into their robotic avatar and then changing the view to the normal view which shows the presentation slides and the lecturer. The second idea was in analogy to a seat reservation system (similar to choosing a seat in the theatre), where remote students see a seating plan of the lecture hall and **remote students “pick their seat” in the lecture hall** (i.e., the Fernstudent). Both approaches could change the mental model of remote students in the desired direction, making them more aware of the situation and their place in the lecture.

We gathered many more feature requests, such as feedback options for remote students to indicate their satisfaction with speed and/or content of the lecture. However, we think that such features should be implemented with care to prevent “feature creep” (i.e., implementing secondary features which do not support the core concept).

In the end, many design decisions depend on the maxim and core intention behind the concept (which could differ from case to case and university to university). We hope, that our experiences and the described method will inspire other researchers and lecturers and are looking forward to collaborations and testing the Fernstudent (or adaptations of it) at other places. For the present case at LMU

Munich, our maxim was to reduce the differences between on-site and remote students, and to create equal rights, responsibilities and interaction opportunities where possible. For example, for the remote students the barrier to participate should be as low as for the on-site students. From that perspective, it makes sense to use natural speech (instead of chat) and the camera being switched on as the default option. In contrast, if the default was total anonymity, the barrier to break up the anonymity and participate in the discussion was much higher. All this reflects a central position in our university: LMU is and will remain a face-to-face university, but it will leverage the experiences from the pandemic and use them wherever they can benefit the future of teaching.

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